AP CALCULUS AB

Unit 1 • Worksheet 1-9A • Answer key

Worksheet 1-9A • IVT, EVT, and MVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

"Was there a time when the water level was exactly 3 m?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) none

Solution

Answer: (A).

Question 2

NO CALCULATOR

"Was the water level ever rising at exactly 0.5 m per hour?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) none

Solution

Answer: (C).

Question 3

NO CALCULATOR

The graph of f on $[0, 4]$ has a corner at $x = 2$ and no breaks. Which theorems are guaranteed to apply on $[0, 4]$?

(A) IVT only
(B) IVT and EVT
(C) EVT and MVT
(D) IVT, EVT and MVT

Solution

Answer: (B). Not differentiable at $x = 2$.

Question 4 (Free response • AP style) NO CALCULATOR

A differentiable function f has the values below.

x	0	2	5	7	10
$f(x)$	4	-2	3	9	5

(a) Must $f(c) = 0$ for some c in $(0, 2)$? Justify.
(b) Must $f'(c) = 3$ for some c in $(5, 7)$? Justify.
(c) Must f attain an absolute minimum on $[0, 10]$? Justify.
(d) Must $f'(c) = 0$ for some c in $(0, 10)$? Justify.

Solution
f is differentiable, so continuous.
(a) Yes, IVT: $f(0) = 4 > 0 > -2 = f(2)$. (b) Yes, MVT: $\frac{9-3}{2} = 3$. (c) Yes, EVT (continuous on a closed interval).
(d) Yes: by the IVT, $f(b) = 4$ for some b in $(5, 7)$; then the MVT on $[0, b]$ gives $f'(c) = \frac{4-4}{b} = 0$.

Question 5 (Free response) NO CALCULATOR

Let $h(x) = x + 1$ for $0 \le x \le 2$ and $h(x) = 3 - \frac{1}{2}(x-2)^2$ for $2 < x \le 4$. Near $x = 2$, the left piece has slope 1 and the right piece has slope 0.

(a) Is h continuous at $x = 2$? Is it differentiable at $x = 2$? Justify.
(b) Which of the IVT, EVT and MVT apply to h on $[0, 4]$? Explain.
(c) Does the MVT apply on $[2, 4]$? If so, what is the guaranteed value of $h'(c)$?

Solution
(a) $h(2) = 3$ and both one-sided limits are 3: continuous. The slopes 1 and 0 differ (a corner): not differentiable at $x = 2$.
(b) IVT and EVT apply (h is continuous on $[0, 4]$). The MVT does not (h is not differentiable at $x = 2$).
(c) Yes: on $[2, 4]$, h is given by $3 - \frac{1}{2}(x-2)^2$, continuous and differentiable. $h'(c) = \frac{h(4)-h(2)}{4-2} = \frac{1-3}{2} = -1$.